

The attacker's model

- **Exhaustive Search**

Try all possible n keys (in average it takes $n/2$ tries)

- **Ciphertext only**

You know one or several random ciphertexts

- **Known plaintext**

You know one or several pairs of random plaintext and their corresponding ciphertexts

- **Chosen plaintext**

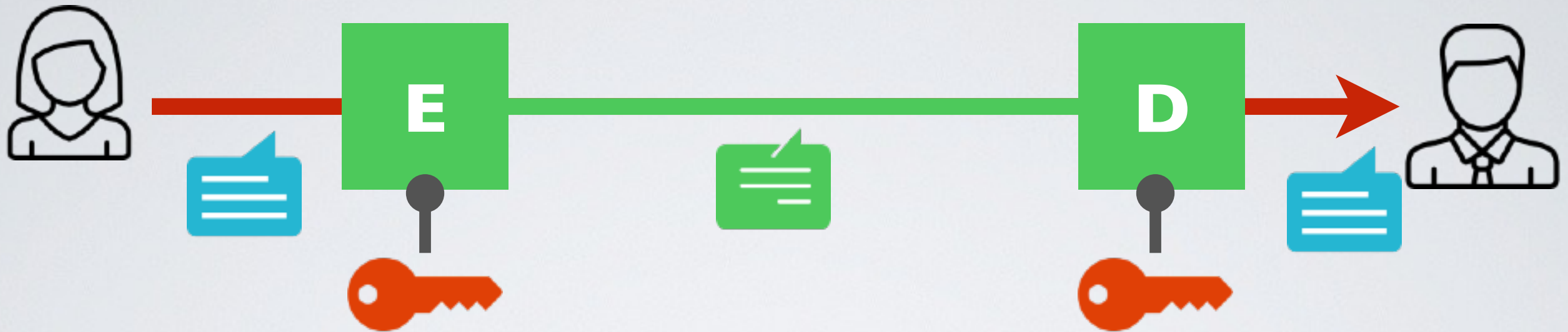
You know one or several pairs of chosen plaintext and their corresponding ciphertexts

- **Chosen ciphertext**

You know one or several pairs of plaintext and their corresponding chosen ciphertexts

➔ **A good crypto system resists all attacks**

Functional Requirements



➡ The same key k is used for encryption E and decryption D

1. $D_k(E_k(m))=m$ for every k , E_k is an injection with inverse D_k
2. $E_k(m)$ is easy to compute (either polynomial or linear)
3. $D_k(c)$ is easy to compute (either polynomial or linear)
4. $c = E_k(m)$ finding m is hard without k (exponential)